MATHEMATICS

MATHEMATICS GRADE 12 TERM 1 CONTROLLED ASSESSMENT PACK

SUBJECT	MATHEMATICS
GRADE	12
ASSESSMENT TASK	MATHEMATICS GRADE 12 TERM 1 CONTROLLED ASSESSMENT PACK
MARKS	50
TIME	1 hour

Educator approval required before classroom use.

INSTRUCTIONS AND INFORMATION

1. Read all instructions carefully.

2. Answer all questions or complete all activities as instructed.

3. Show all working where calculations, planning, diagrams or explanations are required.

4. Use the space provided by the educator, answer booklet or learner task book.

5. Write neatly and clearly.

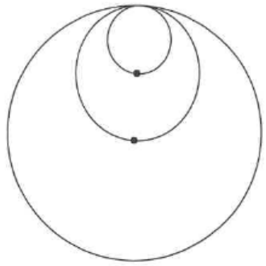
6. Do not use a calculator unless the educator allows it.

SOURCE MATERIAL

SOURCE 1: Official-paper exemplar: Diagram Or Model

QUESTION 3

A circle with radius 6 cm is drawn.
A second, smaller circle is drawn through the centre of the first circle and also touches the first circle internally, as shown in the diagram.
A third, smaller circle is drawn through the centre of the second circle and touches the second circle internally. The process of drawing circles continues and forms a geometric pattern.



3.1 Write down the radius of the 3^{rd} circle. (2)

3.2 Calculate the sum of the areas of the first 10 circles. (4)

3.3 Which circle has a diameter of $\frac{3}{128}$ cm? (4)
[10]

QUESTION 4

Given: $f(x) = a^x - 1$ for $a > 0$. $B\left(2\,;\frac{-5}{9}\right)$ is a point on f.

4.1 Calculate the value of a. (2)

4.2 Write down the range of f. (1)

4.3 Sketch the graph of f. Clearly show the intercepts with the axes and asymptotes, if any. (3)

4.4 It is further given that C is a point on f at $y = \frac{19}{8}$.

Determine the coordinates of C', the image of C, when C is reflected about the line $y = x$. (3)
[9]

SOURCE 2: Official-paper exemplar: Graph Or Chart

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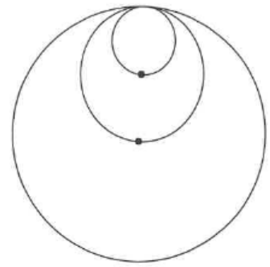

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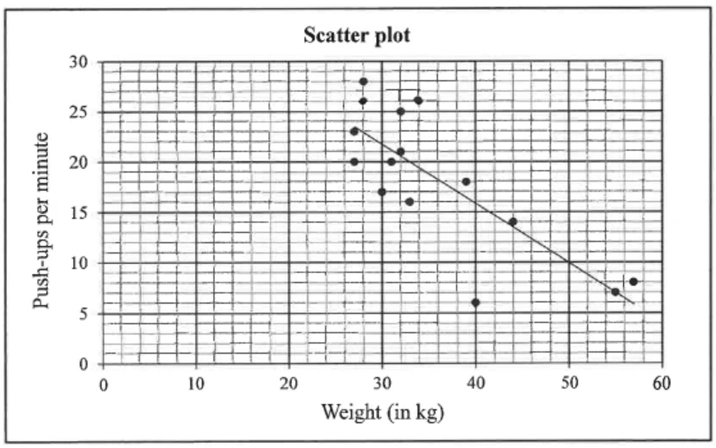
Paper: Mathematics P1 November 2024 | Page: 4 | Source Type: graph_or_chart

SOURCE 3: Official-paper exemplar: Data Table

QUESTION 1

At the beginning of a season, the coach of a junior boys' rugby team recorded the weight (in kg) of the 15 players in his team and the number of push-ups that each player was able to do in one minute. The data is represented in the table and scatter plot below. The least squares regression line for the data is drawn.

Weight (in kg) (x)	34	32	40	27	33	28	27	55	39	44	30	57	28	32	31
Number of push-ups per minute (y)	26	21	6	20	16	26	23	7	18	14	17	8	28	25	20



1.1 Determine the equation of the least squares regression line for the data. (3)

1.2 Write down the correlation coefficient. (1)

1.3 The coach uses the least squares regression line to set the target for the minimum number of push-ups by each team member according to their weight. Predict the number of push-ups that a member of the team, who weighs 29 kg, should do to meet the target. (2)

1.4 Write down the mean number of push-ups for the given data. (1)

1.5 The players trained hard during the season. At the end of the season, the coach reported that each player was able to do 5 more push-ups per minute than they did at the beginning of the season. How does the increase in the number of push-ups influence the standard deviation of the data? (1)

1.6 At the beginning of the season, the coach used the least squares regression line as the minimum target for a player to aim for. Determine the maximum possible increase in the number of push-ups that a team member must obtain to reach the minimum target. (2)
[10]

Paper: Mathematics P2 November 2024 | Page: 3 | Source Type: data_table

QUESTIONS

QUESTION 1: SECTION A: CONCEPTS AND PROCEDURES

Instructions: Show all working.

Stimulus: Topic focus: functions, algebra, equations and financial mathematics.

1.1. Solve a routine problem linked to the stated term focus and show all algebraic working. **(6)**

1.2. Interpret the supplied graph, table or diagram and state the meaning of one key feature. **(6)**

1.3. Complete a multi-step calculation using the supplied values and units. **(8)**

QUESTION 2: SECTION B: PROBLEM SOLVING

Instructions: Use the workspace and justify each step.

Stimulus: Problem context: graphs, transformations, inverses and interpretation of parameters.

2.1. Set up and solve a non-routine problem from the term context. **(10)**

2.2. Justify one method step or proof statement. **(5)**

QUESTION 3: SECTION C: INTEGRATED APPLICATION

Instructions: Combine concepts from the term pack.

Stimulus: Integrated focus: multi-step problem solving with method marks and final answer checks.

3.1. Answer an integrated application question and comment on the reasonableness of the result. **(15)**

MARKING GUIDELINE

TASK 1: SECTION A: CONCEPTS AND PROCEDURES

1.1. Solve a routine problem linked to the stated term focus and show all algebraic working. **(6)**

- Correct method.
- Accurate simplification.
- Final answer stated.

1.2. Interpret the supplied graph, table or diagram and state the meaning of one key feature. **(6)**

- Feature identified.
- Mathematical meaning explained.

1.3. Complete a multi-step calculation using the supplied values and units. **(8)**

- Values substituted.
- Steps logical.
- Answer with unit.

TASK 2: SECTION B: PROBLEM SOLVING

2.1. Set up and solve a non-routine problem from the term context. **(10)**

- Correct representation.
- Valid strategy.
- Accurate conclusion.

2.2. Justify one method step or proof statement. **(5)**

- Relevant theorem/rule.
- Reasoning clear.

TASK 3: SECTION C: INTEGRATED APPLICATION

3.1. Answer an integrated application question and comment on the reasonableness of the result. **(15)**

- Correct plan.
- Accurate working.
- Reasonableness comment.

RUBRIC

Criterion	Marks	Descriptor
Subject accuracy and terminology	12	Uses accurate concepts and terminology.
Evidence, data or source	12	Uses supplied evidence

use		appropriately.
Reasoning and explanation	12	Shows logical, grade-appropriate reasoning.
Communication and completion	14	Answers are clear, complete and structured.

EDUCATOR REVIEW CHECKLIST

- Confirm CAPS term and topic fit before classroom use.
- Insert or approve final source material where the pack requests educator-supplied sources.
- Check mark allocation, memo wording and learner level.
- Confirm visuals are suitable and not treated as official source material unless sourced separately.